# PUNITH B

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## PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate with project experience in machine learning, deep learning, front-end web development, and embedded systems. Worked with Python, scikit-learn, TensorFlow, Keras, HTML, CSS, JavaScript, and Arduino on projects covering cardiac arrhythmia prediction, bone cancer prediction from medical images, an event-planner website, and a dual-axis solar tracker.

## TECHNICAL SKILLS

- **Programming and Web:** Python, HTML, CSS, JavaScript
- **Machine Learning and Deep Learning:** Random Forest, Deep Neural Networks (DNN), VGG16, scikit-learn, TensorFlow, Keras
- **Data Analysis and Visualization:** NumPy, Matplotlib
- **Embedded Systems:** Arduino, Arduino IDE, LDR sensors, servo motors
- **Tools:** Visual Studio Code, Jupyter Notebook

## EXPERIENCE

**Machine Learning Intern | Take It Smart (OPC) Pvt. Ltd.**
**Project:** Cardiac Arrhythmia Prediction Using Random Forest

- Built a machine learning model in Python to predict cardiac arrhythmia.
- Trained the model on datasets imported through scikit-learn.
- Applied the Random Forest algorithm for classification and prediction.
- Used NumPy for numerical computation and Matplotlib for data visualization.

**Sports Analyst | Hudl India**

- Worked as a Sports Analyst for three months.

## PROJECTS

**Bone Cancer Prediction Using Deep Neural Networks (Final-Year Project)**
**Technologies:** Python, TensorFlow, Keras, Jupyter Notebook

- Developed a deep learning model to predict bone cancer from immunohistochemistry (IHC) images.
- Pre-processed IHC images and extracted features using a DNN with the VGG16 model.
- Performed detection by comparing the extracted feature values.

**Dual-Axis Solar Tracker (Mini Project)**
**Technologies:** Embedded Systems, Arduino, Arduino IDE, LDR sensors, servo motors

- Designed a solar panel system that detects the direction of the most intense sunlight and automatically aligns itself towards it.
- Used LDR sensors for light detection and servo motors for two-axis movement, controlled by an Arduino.
- Improved energy capture compared with a fixed, stationary solar panel.

**Event Planner Website (Front-End)**
**Technologies:** HTML, CSS, JavaScript

- Built the front end of a website for an event planner using HTML, CSS, and JavaScript.
- Live site: https://34dexter.github.io/Eventrra._/

## EDUCATION

**Bachelor of Engineering (BE), Electronics and Communication Engineering**
Rajarajeswari College of Engineering, Bangalore | 2023

## LANGUAGES

English, Kannada, Hindi